



QUALCOMM | SNAPDRAGON® AI LAB BUILD & PRESENT CHALLENGE

SmartQC: On-Device Visual Quality Inspection

Optimized for Snapdragon

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THE PROBLEM

Quality inspection is slow, inconsistent, or expensive

1

Manual inspection

Depends on human attention span — fatigue and inconsistency creep in over a shift.

2

Cloud-based AI tools

Solve accuracy, but add latency, an internet dependency, and data-privacy exposure.

3

Industrial inspection hardware

Accurate and fast, but priced far out of reach for small food producers and workshops.

There is a need for a fast, private, self-contained inspection tool that runs entirely on the device performing the inspection.

THE SOLUTION

SmartQC: factory-grade inspection, without the factory-grade price tag

WHO IT'S FOR

Small bakeries, food producers, and workshops that need reliable visual QC but can't justify a dedicated inspection line.

WHAT IT DOES

Classifies product quality from a webcam or uploaded image, entirely on-device — no cloud calls, no internet dependency at inference time.

WHY IT MATTERS

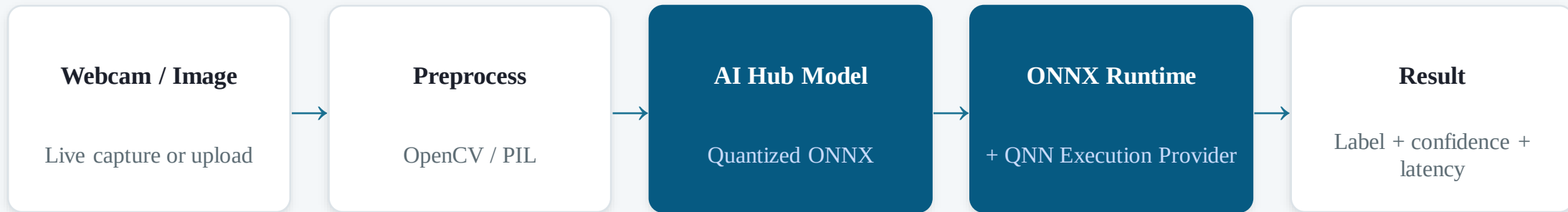
Runs on a Snapdragon laptop the business may already own — zero extra hardware cost.

VALUE PROPOSITION

“Factory-grade visual inspection, without the factory-grade price tag.”

HOW IT WORKS

Architecture



The optimized model and ONNX Runtime + QNN Execution Provider path are what make this pipeline Snapdragon-NPU-ready.

App features: live webcam / upload inspection • batch folder inspection + CSV export • confidence threshold flagging
• local SQLite history log • rolling defect-rate trend chart • fully offline at inference time

WHY ON-DEVICE, WHY SNAPDRAGON

Built for where inspection actually happens

Cloud-based inspection

- ✗ Needs constant internet
- ✗ Per-image / API cost
- ✗ Data leaves the device
- ✗ Latency varies with connection

SmartQC on Snapdragon

- ✓ Fully offline at inference
- ✓ Zero marginal cost per scan
- ✓ All data stays on-device
- ✓ NPU-optimized, low-latency

Snapdragon's on-device NPU + long battery life means SmartQC can run all day on a shop floor — unplugged, private, and instant.

MODEL TRAINING

MobileNetV2, fine-tuned to 99.2% validation accuracy

99.2%

Best validation accuracy

649

Total labeled images (Fresh / Mold bread)

12

Training epochs

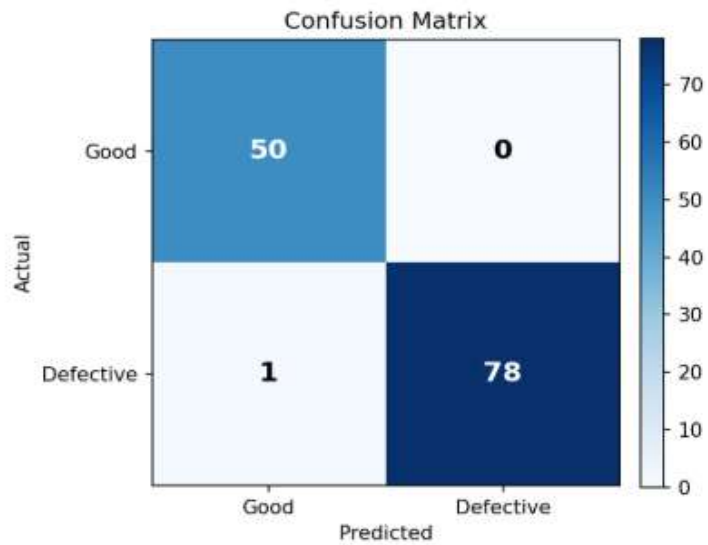
```
● (venv) PS C:\Users\samay\OneDrive\Desktop\SmartQC\app> cd ..
● (venv) PS C:\Users\samay\OneDrive\Desktop\SmartQC> python train.py
Found classes: ['Fresh Bread(Good Bread) Images', 'Mold Bread(Bad Bread) Images']
Total images: 649
Train: 520 Val: 129
Epoch 1/12 train_loss=0.2621 train_acc=0.902 val_acc=0.984
Epoch 2/12 train_loss=0.1234 train_acc=0.956 val_acc=0.992
Epoch 3/12 train_loss=0.0647 train_acc=0.977 val_acc=0.992
Epoch 4/12 train_loss=0.0595 train_acc=0.981 val_acc=0.984
Epoch 5/12 train_loss=0.0374 train_acc=0.992 val_acc=0.984
Epoch 6/12 train_loss=0.0208 train_acc=0.994 val_acc=0.984
Epoch 7/12 train_loss=0.0169 train_acc=0.994 val_acc=0.977
Epoch 8/12 train_loss=0.0135 train_acc=0.996 val_acc=0.984
Epoch 9/12 train_loss=0.0538 train_acc=0.992 val_acc=0.984
Epoch 10/12 train_loss=0.0698 train_acc=0.977 val_acc=0.984
Epoch 11/12 train_loss=0.0220 train_acc=0.990 val_acc=0.969
Epoch 12/12 train_loss=0.0202 train_acc=0.996 val_acc=0.984

Best validation accuracy: 0.992
```

Fine-tuned on a Fresh Bread (Good) / Mold Bread (Defective) dataset using transfer learning from ImageNet-pretrained MobileNetV2

MODEL EVALUATION

Beyond accuracy: confusion matrix, precision/recall, ROC-AUC

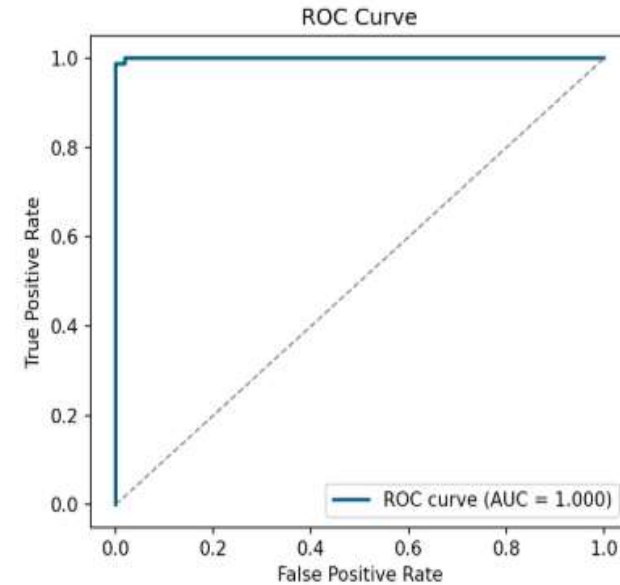


129

Validation images

1

Total misclassifications



0.9997

ROC-AUC

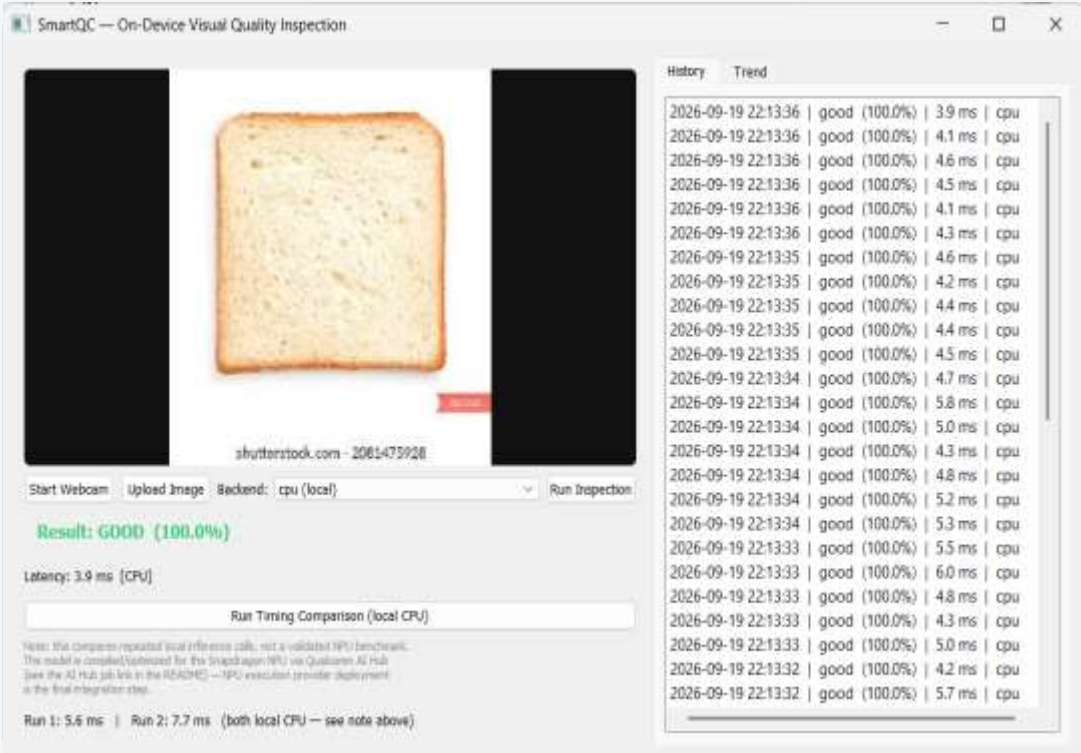
Compiled and optimized for the Snapdragon X Elite

COMPILE QUANTIZE LINK PROFILE INFERENCE						
Creator: samaythakur999@gmail.com (me) ▾		Rows per page: 10 ▾ 1-3 of 3 < < > >				
Job Name	Job ID	Status	Target Device	Creator	Submission Time	
smartqc_modelonnx	jpe782z15	✓ Results Ready	Snapdragon X Elite CRD	samaythakur999@gmail.com	9/20/2026, 2:51:55 AM	

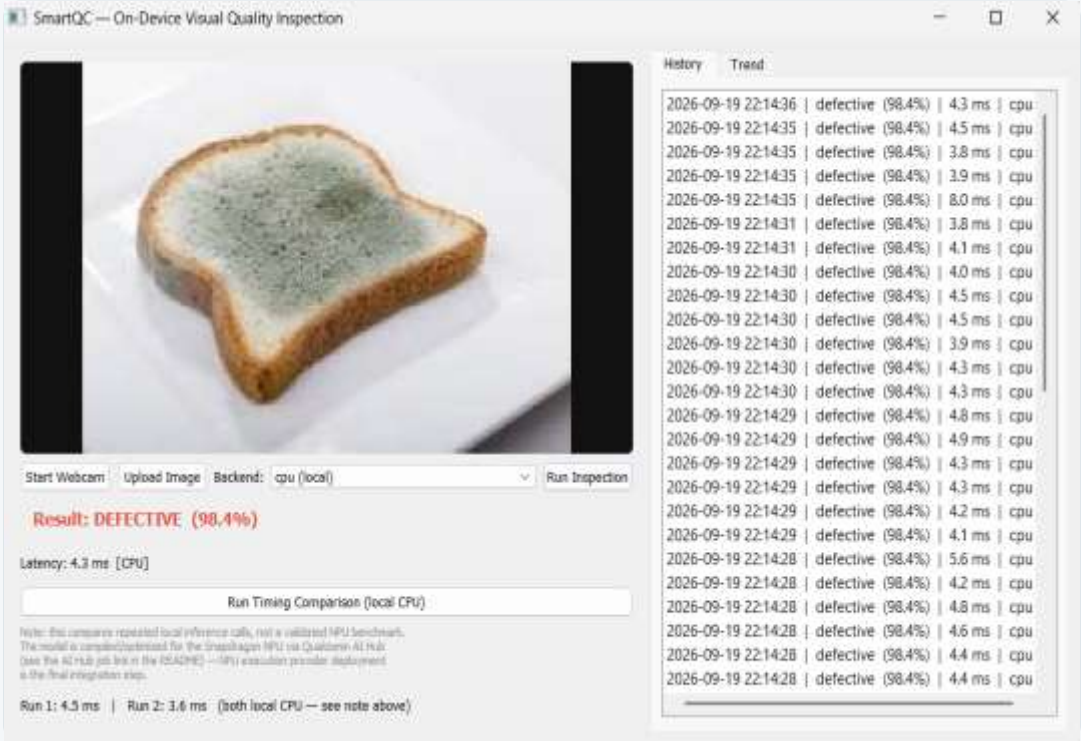
Compile job jpe782z15 • Target: Snapdragon X Elite CRD • Status: Results Ready

LIVE DEMO

Correctly classifies both good and defective samples



GOOD — 100.0% confidence

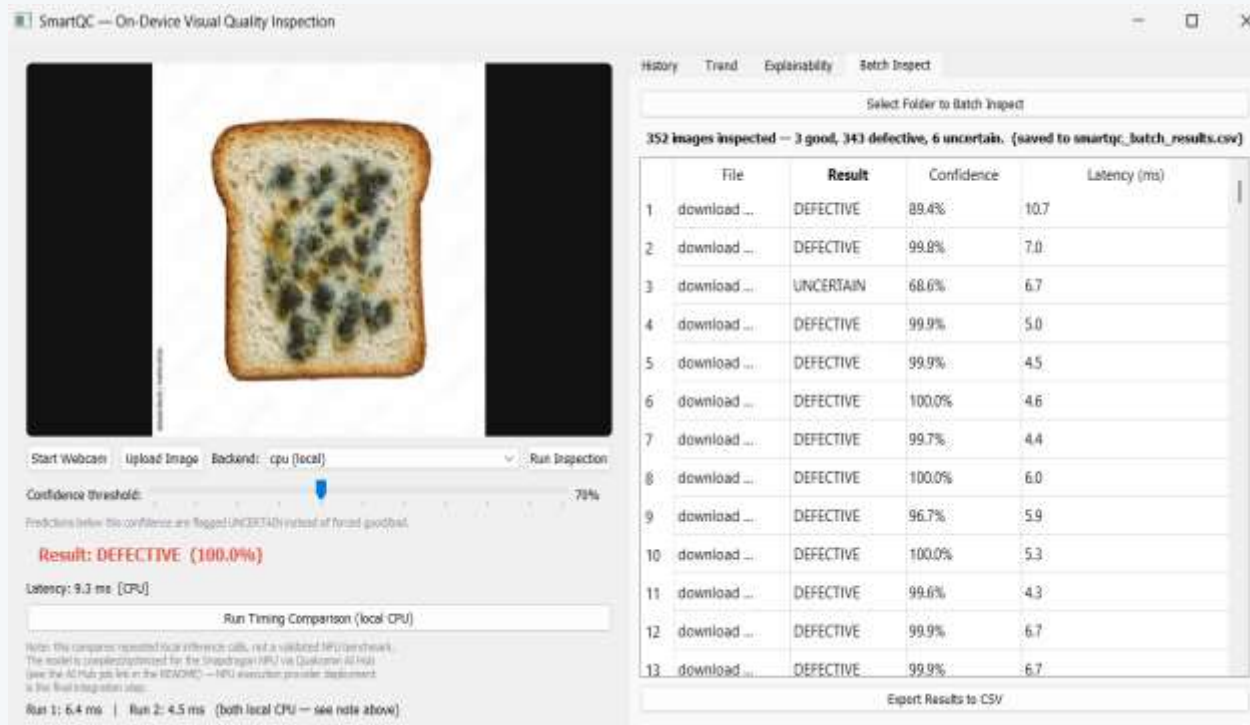


DEFECTIVE — 98.4% confidence

REAL-WORLD VALIDATION

Batch-tested against 352 uncurated images

Beyond the curated 129-image validation set, SmartQC's batch mode was run against a larger, unsorted folder of 352 real-world mold-bread images — a messier, more realistic test than a clean benchmark.



352

Images batch-processed

343

Correctly flagged
DEFECTIVE

6

Flagged UNCERTAIN
(below threshold)

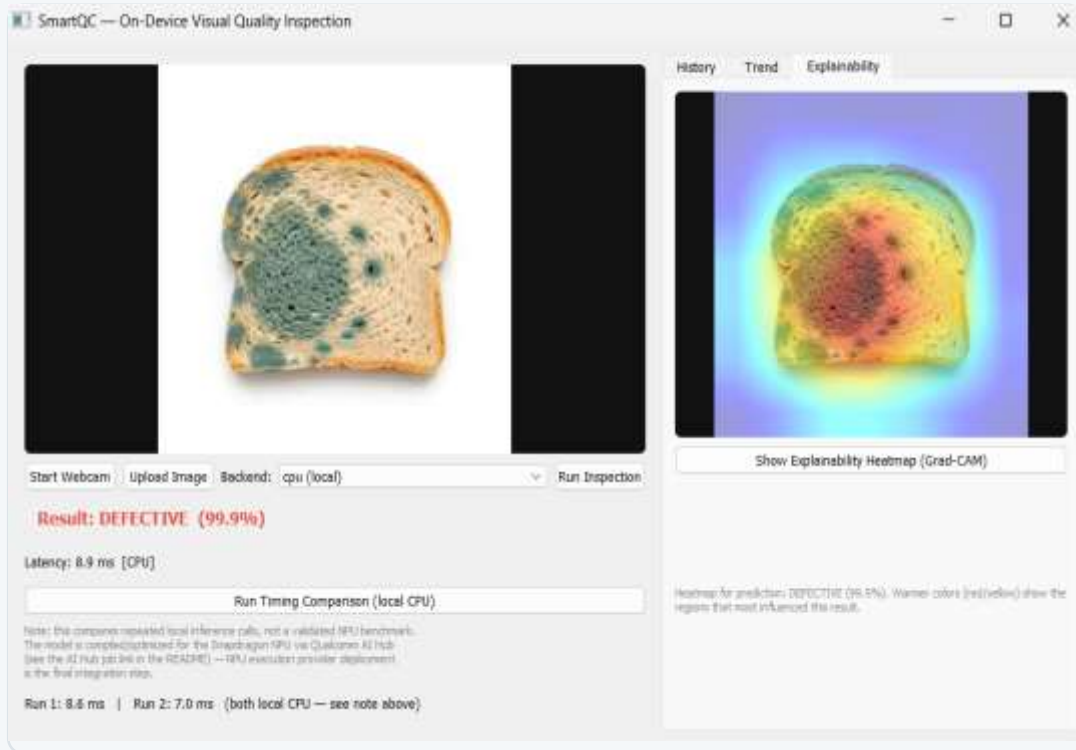
3

Misclassified (edge cases)

97.4% correct on an uncurated real-world batch — and the confidence-threshold feature caught the ambiguous cases instead of silently guessing wrong.

BEYOND CLASSIFICATION

Explainability: showing WHY, not just what



Grad-CAM heatmap

Highlights exactly which pixels drove the model's decision — warmer colors (red/yellow) show the highest-influence regions.

Builds operator trust

A QC operator can visually verify the model is looking at the actual defect, not an unrelated artifact.

Runs on the trained model directly

Computed via PyTorch alongside the ONNX/AI-Hub inference path — an explainability layer, not a replacement for it.

WHAT'S NEXT

Roadmap

1

Multi-product support

Extend beyond bread to additional inspection domains (produce, packaging, components).

2

Multi-class defects

Move beyond good/bad to specific defect types — mold, staleness, physical damage.

3

Fleet / multi-station rollout

Central dashboard aggregating results across multiple inspection stations.



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Thank You